

Content Inventory and RAG Addendum Report

This pass completed the curriculum inventory artifacts, added RAG and Retrieval as a dedicated Journey card, added Grounding to the glossary, upgraded the lesson-card review route, exported the review PDF, and verified the app builds.

Journey count

26 lessons inventoried. RAG and Retrieval is lesson 22, immediately after Context Window.

PDF artifacts

`docs/content-inventory/prompt-life-lesson-cards-v0-6.pdf` and this report PDF. The review deck has 27 PDF page objects: one cover plus 26 lessons.

Verification

`npm run typecheck`, `npm run build`, and `npm run export:lesson-cards` passed.

Deploy target

Main branch update prepared for GitHub Pages phone testing.

Files Changed

- `src/data/content.ts`, `src/data/exercises.ts`, `src/data/contentReview.js`
- `src/main.tsx`, `src/components/VisualAids.tsx`, `src/styles/global.css`
- `scripts/export-lesson-pdf.mjs`, `scripts/generate-content-inventory.mjs`, `package.json`, `README.md`
- `docs/content-inventory/*`, `docs/REVIEW_NOTES.md`, and `docs/screenshots/v0-6-3-*.png`

What Is Done

- Generated complete current-content inventory, quality rubric, lesson matrix JSON, gaps, architecture proposal, visual inventory, misconception map, rewrite draft, UI copy cleanup list, and review instructions.
- Added RAG and Retrieval / Open-Book AI as a Journey card with prompt/response and durable/temporary distinctions.
- Added Grounding glossary content with metaphor, Brain Bridge, and limitation.
- Enhanced `/review/lesson-cards` with inventory fields and rubric scores.
- Exported the inventory lesson-card PDF with one review card per lesson.

How To Run Locally

- `npm install`
- `npm run dev`
- Open <http://localhost:5173/>
- For the review route, open <http://localhost:5173/review/lesson-cards>

Next Three v0.2 Improvements

- Rewrite the highest-priority lessons in app data: Vectors, Tensors, Attention, MLP, Hidden States, Context Window, RAG, and Risk vs Myth.
- Add a shared dog/cat example path across tokenization, attention, decoding, autoregression, context, and RAG.
- Add automated route and mobile smoke tests for lesson count, RAG placement, PDF export, bottom-nav clipping, and glossary drawer fields.

Lesson Cards

One accessible curriculum-review card per lesson, built from current app data plus inventory notes and rubric scores.

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BEFORE MORNING

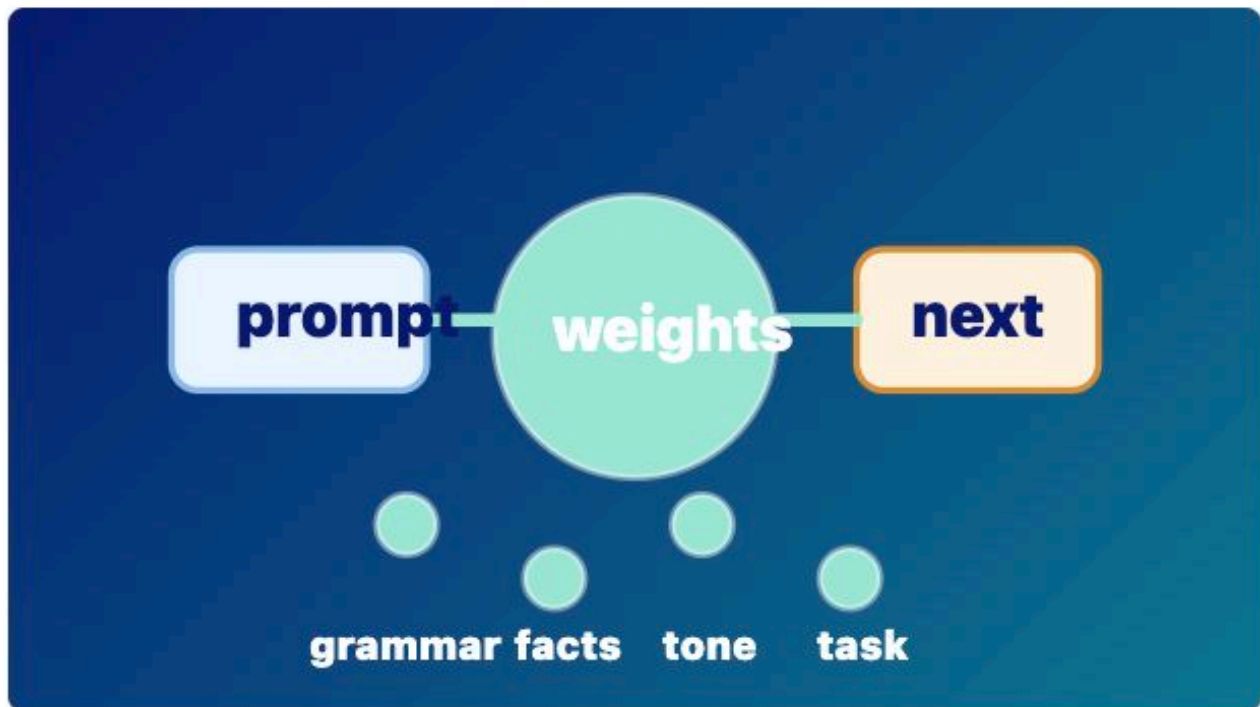
What Is an LLM?

A learned next-token prediction system

Stage: architecture

Priority: low

Rubric avg: 4.4/5



Prompt to Prediction

A prompt enters the current context; learned weights shape next-token probabilities.

DEFINITION

A large language model is a learned system that predicts likely next tokens from the context it can see.

ILLUSTRATION NEEDED

Limited stack of cards where older cards fall out.

Rubric Scores

Accuracy	5/5	Placement	5/5
Relationship	5/5	Prompt/response	5/5
Durable/temp	5/5	Metaphor	4/5
Brain/cognition	4/5	Visual	4/5
Exercise	4/5	Mobile	4/5

context window

input context

prompt tokens

response tokens

rag

autoregression

inference

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THE DAY REPEATS

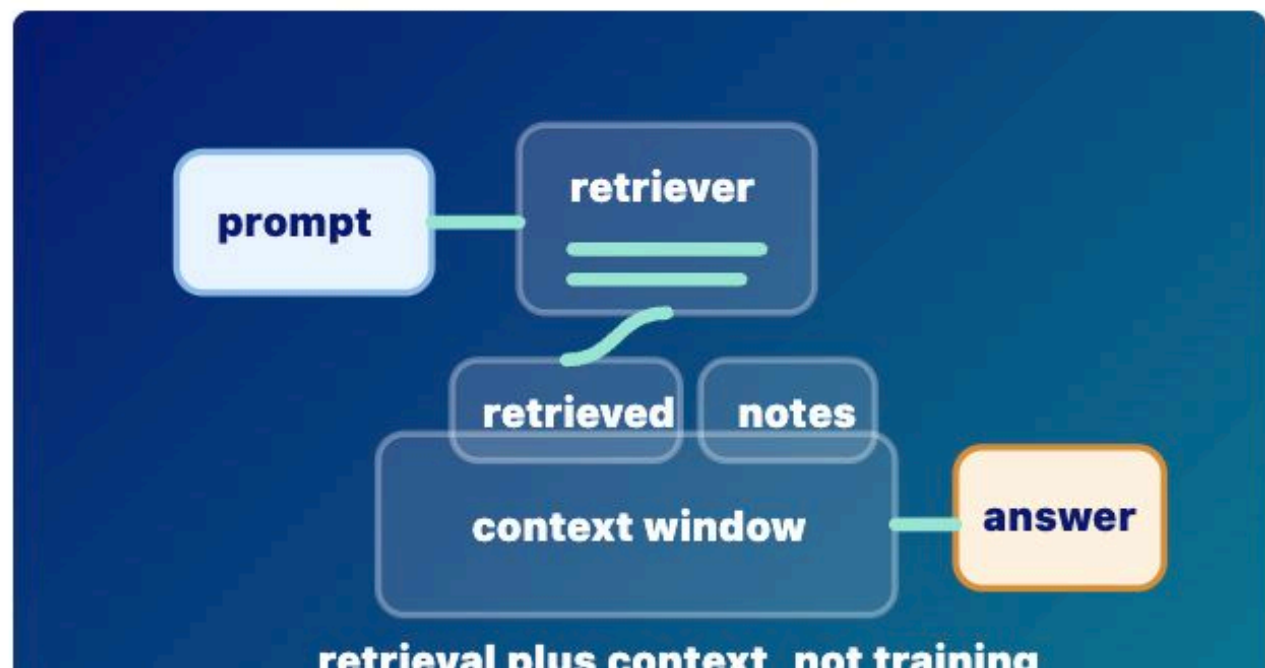
RAG and Retrieval

Retrieved text joins the current context

Stage: current context

Priority: high

Rubric avg: 4.6/5



RAG and Retrieval review card, including the dedicated visual aid.

RAG and Retrieval

Retrieved text joins the current context

Retrieval-augmented generation, or RAG, retrieves outside information and places it into the model's context before generating a response.

RAG

Retrieval

Grounding

Context window

Input context

Prompt tokens

Response tokens

Inference

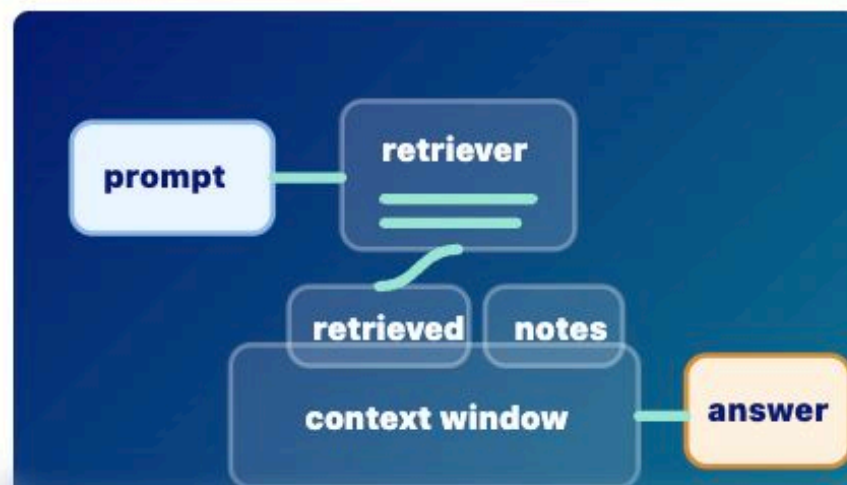
Fine-tuning

Training

Hallucination

VISUAL AID

What to picture



Home

Journey

Play

Glossary

Badge

1

Before
Morning

2

Morning
Commute

3

Workday

4

Decision
Room

5

The Day
Repeats

6

Wider AI
Literacy

RAG and Retrieval

GLOSSARY

[Close](#)

Grounding

Connecting a model response to available evidence, such as retrieved documents, cited sources, data, or tool results.

Relationship

RAG is one way to improve grounding, but grounding can still fail if retrieval is poor or the model misuses the retrieved context.

Metaphor

Tying the balloon to the ground.

Brain bridge

Like checking your answer against notes before speaking.

Where it breaks

The model is not consciously verifying truth unless the system is designed to check evidence.

Often confused with

Grounding can help, but it does not make every answer true.